

Advancing Alzheimer's Therapy: Nanomedicine, Cell-Specific Drug Discovery, and Anti-Amyloid Strategies

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INTRODUCTION

Alzheimer's Disease (AD) is a neurodegenerative disorder, which impacts the human brain, leading to damage in the brain memory. Over 50 million people are currently living with AD or other related dementias worldwide, a number that is projected to triple by 2050 (Blumenfeld 2025). AD is the most common type of dementia, resulting in tens of millions of elderly people suffering from cognitive decline and behavioral impairment (Zhao 2023).

AD accounts for almost 70% of dementia cases, and its pathophysiology is characterized by an accumulation of small peptides, amyloid- β ($A\beta$), in fibrils and plaques, followed by hyperphosphorylation, misfolding, and aggregation into neurofibrillary tangles of another protein, tau. Both aggregates are associated with strong inflammatory responses, synaptic dysfunction, and neuronal injury, causing considerable brain damage and impairing cognitive processes.

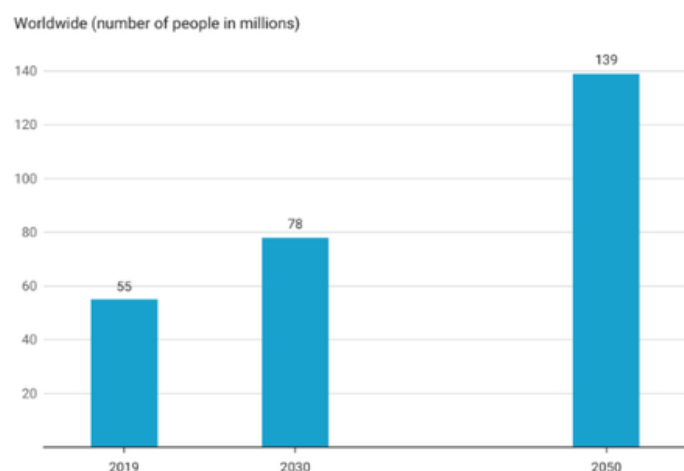


Figure 1: Number of people (by millions) living with AD. Numbers are projected to triple from 2019 to 2050 (Media.us Market).

The blood-brain barrier (BBB), a selective permeability barrier, which protects the central nervous system (CNS) and safeguards it from potentially harmful substances. The dysfunction of BBB leads to the accumulation of amyloid-beta ($A\beta$) plaques and neurofibrillary tangles, which leads to the development of Alzheimer's Disease.

This review focuses on Alzheimer's therapies, including nanoparticle-mediated BBB repair, cell-type-specific drug discovery and anti-amyloid approaches, highlighting how they target key AD mechanisms.

HYPOTHESIS

Analyzing current research on nanoparticle-mediated cure, cell-type-specific drug discovery and anti-amyloid approaches will clarify how these strategies affect and target current AD mechanisms and inform future therapeutic directions.

ALZHEIMER'S DISEASE

A cell-type-specific, multi-target drug discovery strategy is introduced, grounded in human data and real-world evidence. This approach integrates single-cell *transcriptomics*, drug *perturbation* databases, and clinical records. Using this framework, *letrozole* and *irinotecan* are identified as a potential combination therapy, each targeting AD-related gene expression changes in neurons and *glial* cells, respectively. In an AD mouse model with both $A\beta$ and *tau* deposits, this combination therapy is significantly improving memory performance and reducing AD-related pathologies compared with vehicle and single-drug treatments. Single-nucleus transcriptomic analysis confirmed that

the therapy reversed disease-associating gene networks in a cell-type-specific manner (Cell 2025)

BLOOD-BRAIN BARRIER

The downregulation of Low-Density Lipoprotein Receptor-Related Protein 1 (LRP1) is strongly correlated with impairment of the BBB and cognitive decline. Proper regulation of LRP1 levels in endothelial cells is crucial for preventing the progression of AD. Despite this, the mechanisms that maintain appropriate LRP1 levels on the basolateral surface of endothelial cells remain unclear (Chen et al. 2025)

NANOTECHNOLOGY

Nanotechnology (NT) enables a new, alternative pathway for development of AD treatment interventions. At present, the NT treatments for attenuation of AD memory impairment are at the animal model stage. Over the past four years, there has been a steady increase in publications of AD animal models with a wide variety of original NT treatment interventions, able to attenuate memory impairment. NT therapy development, in animal models of AD, is faced with the twin challenges of the nature of AD, a chronic impairment, unique to humans, of the tau protein and A β peptides that regulate several key physiological brain processes, and the incomplete understanding of AD's aetiology. This paper reviews the state-of-the-art in NT based treatments for AD memory impairment in animal models and discusses the future work for translation to the successful treatment of AD cognitive impairment in human. Exploration in A β -Targeted Preclinical Nanomedicine (Samo Ribarič 2022)

EXPLORATION OF AB-TARGETED PRECLINICAL NANOMEDICINE

The intervention of A β production from the source seems to be a promising way to decrease the intracerebral A β . There are two

main enzymes, beta-site amyloid precursor protein cleaving enzyme 1 (BACE-1) following γ -secretase enzyme, cleave the transmembrane APP and achieve amyloidogenic cleavage. (Zhao et al. 2023).

A study demonstrates that using mid-affinity, multivalent LRP1-targeted nanoparticles (specifically angiopep-2-conjugated polymersomes) can rapidly clear about 41% of brain amyloid- β (A β) deposits in Alzheimer's disease mouse models within hours, restore blood-brain barrier (BBB) function, and result in long-lasting cognitive recovery, as measured by behavioral tests.

ANTI-AMYLOID-B

An international study by WashU in 2025 studied 73 people indicated to carry a rare inherited genetic mutation that causes the overproduction of amyloid in the brain, all but guaranteeing that they will develop Alzheimer's disease in middle age. They evaluated effects of an affected anti-amyloid drug to see if the medication could prevent the development of dementia. (Tamara Schneider, 2025) Participants received Lecanemab - an anti-amyloid drug to slow cognitive decline in people who already have symptoms of Alzheimer's Disease.

LECANEMAB AND DONANEMAB

Lecanemab, a humanized IgG1 antibody derived from the mouse antibody mAb158, received accelerated approval by FDA in January 2023 and full approval in July 2023 for the treatment of mild cognitive impairment (MCI) or mild dementia associated with AD (Zhao et al. 2023)

Lecanemab targets a protein called amyloid that builds up in the brains of people with Alzheimer's disease. It is an immunotherapy, which means it uses the immune system of the brain to remove the amyloid protein. This has been shown to slow down the decline in memory and thinking skills of people with early Alzheimer's disease (Alzheimer's Society 2025).

In the TRAILBLAZER-ALZ trial, Donanemab treatment led to a 32% slower decline on the Integrated Alzheimer's Disease Rating Scale over 76 weeks, with significant amyloid plaque reduction observed in patients (Yadri 2025).

Donanemab works in a similar way to lecanemab. The final recommendation from the National Institute for Health and Care Excellence (NICE) is that donanemab will not be made available on the NHS (Alzheimer's Society).

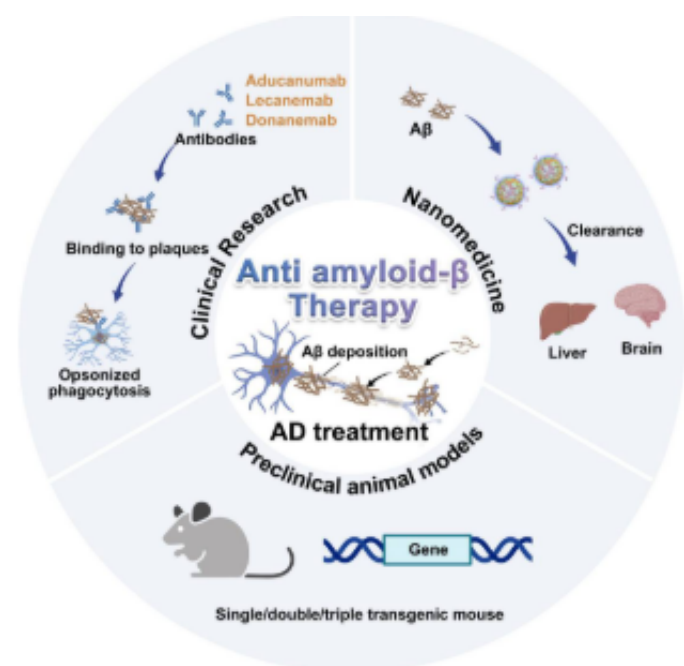


Figure 2: The three main directions of AD treatment research progression (Zhao et. al. 2023).

RESEARCHING NEW DRUG TREATMENTS

Semaglutide is a medication primarily known for its use in managing type 2 diabetes and more recently, aiding in weight loss. It is made by Novo Nordisk and sold as a pill and marketed under the brand names Ozempic and Wegovy. Research into people taking drugs like semaglutide for diabetes showed that they had a reduced risk of developing dementia. This led to clinical trials to test the potential role of semaglutide in helping to treat Alzheimer's disease.

The latest drug development focuses on disease-modifying approaches, especially anti-

amyloid monoclonal antibodies (like lecanemab and donanemab) but highlights ongoing challenges of efficacy, safety, high costs, and the necessity for multi-targeted treatments due to Alzheimer's complex, multifactorial nature.

CONCLUSION

Alzheimer's disease is incredibly complicated because so many things go wrong at the same time: the buildup of amyloid- β , changes in tau proteins, inflammation in the brain, and even problems with the blood-brain barrier. When key transport proteins in the BBB like LRP1 stop working properly, amyloid- β piles up faster, which can speed up memory loss.

New nanotechnology-based treatments seem to help repair the BBB, improve how drugs get into the brain, and even reduce the amount of amyloid. Anti-amyloid treatments like Lecanemab and Donanemab are big steps forward in Alzheimer's therapy. They slow down cognitive decline in people who catch the disease early, but the improvements are still limited, and things like the high cost, limited availability, and side effects such as ARIA make them hard to use widely. There are also newer approaches that try to stop amyloid from being produced in the first place by targeting enzymes like BACE-1 and γ -secretase but these need to be handled carefully so they don't mess with important normal functions in the body.

Overall, Alzheimer's treatment is moving toward a more multi-targeted, personalized style mixing nanomedicine, cell-specific treatments, immunotherapy, and biomarker-based methods. We don't have a cure yet, but progress in early diagnosis, understanding the blood-brain barrier, and improving drug delivery is helping scientists slowly change how this disease develops. Moving forward, combining different approaches and making treatments both safer and more accessible will be key, along with getting a clearer picture of the molecular changes that actually kick-start and drive Alzheimer's.